Population and Economic Growth
When Malthus Meets Romer

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One of the Oldest Debates in Economics

- Does a larger population raise or lower living standards?
  - → Malthus (1798): population strains resources ⇒ wages to subsistence
    - ∗ David Ricardo, Paul Ehrlich, William Catton, . . .
  - → Romer (1990): more people ⇒ more ideas ⇒ innovation dominates
    - ∗ Richard Jones, Julian Simon, Chad Jones, Elon Musk, . . .
- Never more consequential: 2bn to 8bn+ in a century, now fertility below replacement across the developed world
  - → "The end of economic growth" – Jones (2022)
- This paper: study the implications of the disagreement itself

This Paper

- Overlapping generations model with
  - → Fertility choice under altruism
  - → Semi-endogenous growth dynamics
  - → Heterogeneous beliefs about how population matters for productivity
- Mechanism: beliefs affect fertility choice & fertility shapes evidence ⇒ wrong beliefs can persist and delay growth
- For mechanism to work, only need
  1. Population matters for productivity (core of growth models)
  2. Children's living standard matters for fertility choice (core of intergenerational models)

Findings

- Theoretical predictions
  - → Fertility ranking between types is state-dependent
  - → Wrong beliefs generate Malthusian cycles and Romerian traps
- Supporting evidence
  - → Belief correlates with fertility in survey data
  - → Complementary evidence from Malthus at Haileybury
- Quantitative assessment
  - → Inherited disagreement delays takeoff by centuries
  - → Information trap if initial Romerian share too small

Related Literature & Contribution

- **Fertility & unified growth**: Barro–Becker (1989), Galor (2000, 2011), Hansen–Prescott (2002); demographic transition and sustained growth
- **Scale effects**: Romer (1990), Kremer (1993), Jones (1995, 2022); the sign of the population–technology link is central to long-run projections
- **Heterogeneous beliefs**: Angeletos–La'O (2013), Coibion–Gorodnichenko (2015); climate disagreement (Bakkensen–Barrage 2022)
- **Cultural transmission/entrepreneur & evolutionary selection**: Bisin–Verdier (2001, 2011), Galor–Moav (2002), Galor–Michalopoulos (2012), Mokyr (2016)

This paper: disagreement about the sign of the population scale effect, where fertility choices shape both the demographic composition of beliefs and the evidence available to evaluate them

Roadmap

- Model
- Supporting evidence
- Equilibrium dynamics
- Application with quantitative illustration

Model Setup

Model: Environment

- Discrete-time OLG; continuum of atomistic households
- Time endowment 1; fertility n costs time $\psi(n)$, labor supply $1 - \psi(n)$
- Linear production: $Y_t = A_t L_t$, wage $= A_t$, so

$$c_{j,t} = A_t\big(1 - \psi(n_{j,t})\big)$$

- Population and its adjusting reference:

$$N_{t+1} = \bar{n}_t N_t, \qquad \bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}$$

- ϕ_t = share of Romerians at time t

TFP Dynamics: The Population Gap

microfoundation

- Productivity growth:

$$\ln A_{t+1} - \ln A_t = \theta^* x_t + \eta_{t+1}, \qquad \eta_{t+1} \sim \mathcal{N}(0, \sigma^2)$$

- Population gap x_t:

$$x_{t+1} = (1 - \mu) x_t + \ln \bar{n}_t, \qquad \mu \in (0, 1]$$

- μ = catch-up rate of the reference (semi-endogenous, Jones 1995):
  - → one-time population increase ⇒ temporary growth effect
  - → sustained growth $\bar{n} > 1$ ⇒ permanent: $g_\infty = \theta^* \ln \bar{n} / \mu$
- Disagreement: $\theta^* \in \{\theta_R, \theta_M\}$, $\quad \theta_M < 0 < \theta_R$ but unknown to households

Preferences: Barro–Becker Dynasties

why these preferences

- Dynastic utility:

$$V_{jt} = \max_{n_{jt}} \; u\big(c_{jt}\big) + \beta\, n_{jt}^{1-\varepsilon}\, \mathbb{E}_j\big[V_{j',t+1}\big], \qquad \rho, \varepsilon \in (0,1)$$

  subject to $c_{jt} = A_t(1 - \psi(n_{jt}))$

- Time cost of childbearing: increasing and convex
- CRRA utility + time costs $\Rightarrow$ TFP level cancels from the fertility FOC
- Beliefs matter only via expected growth

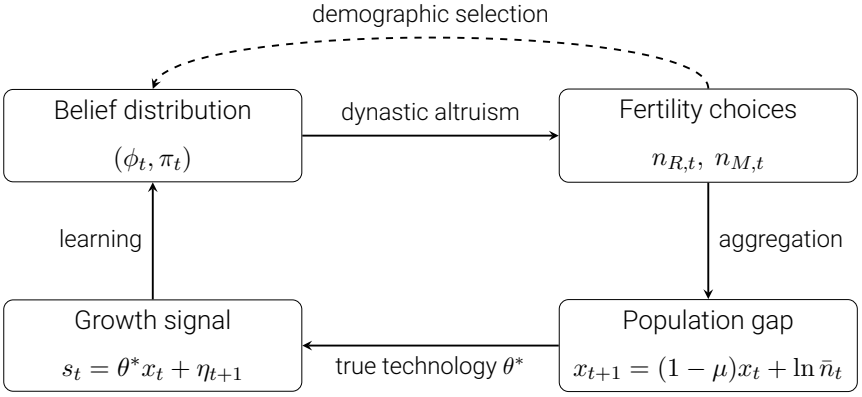
Beliefs, Learning, and Transmission

- Two types: share ϕ_t Romerians ($\theta_R > 0$), $1 - \phi_t$ Malthusians ($\theta_M < 0$)
- Social credence π_t: shared posterior (the body of knowledge in the society), Bayes-updated from growth signals $s_t = \theta^* x_t + \eta_{t+1}$
- Transmission: child inherits parent's worldview with prob λ; otherwise adopts the Romerian view with prob π_{t+1}
- Belief distribution evolves via two channels:

$$\phi_{t+1} = \underbrace{\lambda\, \phi_t^{birth}}_{\text{cultural transmission}} + \underbrace{(1-\lambda)\, \pi_{t+1}}_{\text{learning-based recruitment}}$$

- Reproductive selection and social learning can conflict

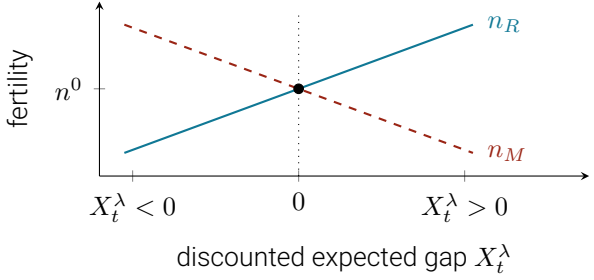
The Belief–Fertility–Growth Feedback Loop



- Near $x = 0$: both feedback channels into beliefs shut down together
- Household choices shape the evidence that future generations use to revise beliefs

State-Dependent Fertility Ranking

$$n_{R,t} - n_{M,t} \approx \underbrace{\frac{1-\rho}{1+\varepsilon}}_{K} \cdot \Delta\theta \cdot X_t^{\lambda} \quad \text{where} \quad X_t^{\lambda} \equiv \mathbb{E}_t\left[\sum_{k\geq 0}(\delta\lambda)^k x_{t+k}\right]$$



- $X_t^{\lambda} > 0$ (above reference on balance): Romerians out-reproduce
- $X_t^{\lambda} < 0$ (below reference on balance): ranking flips
- At $X_t^{\lambda} = 0$: identical fertility despite disagreement

Empirical Support

Beliefs and Fertility: GSS Evidence

by education completed fertility

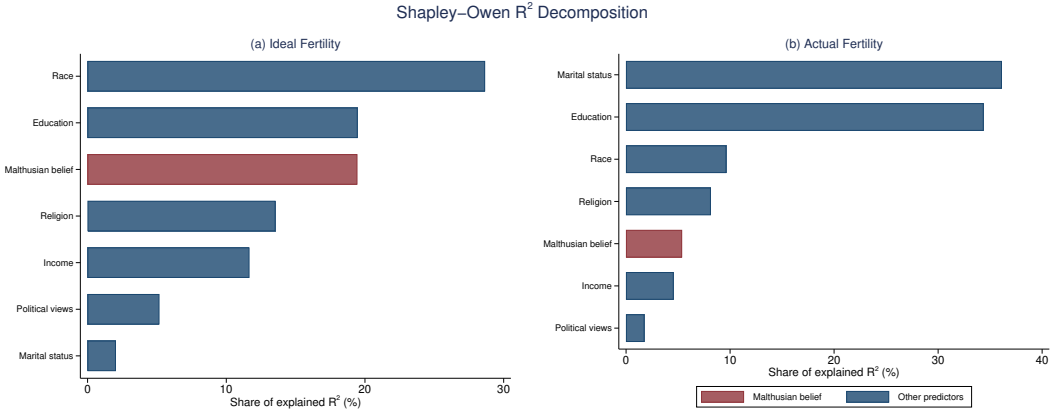
- GSS 2000 & 2010 ($N = 1{,}822$): agree/disagree "Earth cannot continue to support population growth" — 55% Malthusian

	Ideal Fertility		Actual Fertility	
	Baseline	Full	Baseline	Full
Malthusian	-0.268^{***}	-0.203^{***}	-0.292^{***}	-0.237^{***}
	(0.041)	(0.045)	(0.066)	(0.071)
Demographic controls + Year FE	Yes	Yes	Yes	Yes
Additional controls	No	Yes	No	Yes
Observations	1,822	1,567	1,822	1,567
R^2	0.060	0.099	0.262	0.333

- Baseline: age, age^2, sex, education, year FE
- Full adds: marital status, race, religion, political views, income

Explanatory Power

compare



Extensive Margin and Reverse Causality

details age groups

Margin decomposition:

- Extensive: Malthusian odds ratio = 0.55 (AME: -9.2 pp, $p < 0.001$)
- Intensive (conditional on > 0): -0.12 ($p = 0.089$)
- Association concentrated on the decision to become a parent

Reverse causality checks:

- Ideal-fertility gradient significant in all age groups, including 18–34 (limited childbearing experience)
- Actual-fertility: zero among young, large among 35+
- Siblings control: coefficient changes $<5\%$

- Adding general environmental concern does not affect the results
- Pattern more consistent with beliefs $\rightarrow$ fertility

Complementary Evidence: Malthus at Haileybury

- Robertson (2026): British Indian officials trained at Haileybury, where Malthus himself taught political economy
- Malthus's abrupt death $\Rightarrow$ replaced by Richard Jones (a critic) $\Rightarrow$ quasi-exogenous variation in doctrine across cohorts
- Officials taught by Malthus provided systematically less famine relief
- Three features: exogenous assignment, costly behavior (not cheap talk), decades-long durability
- Complements GSS: clean identification of the belief channel, though in bureaucratic relief rather than fertility

Equilibrium Dynamics

Demographic Selection and Learning

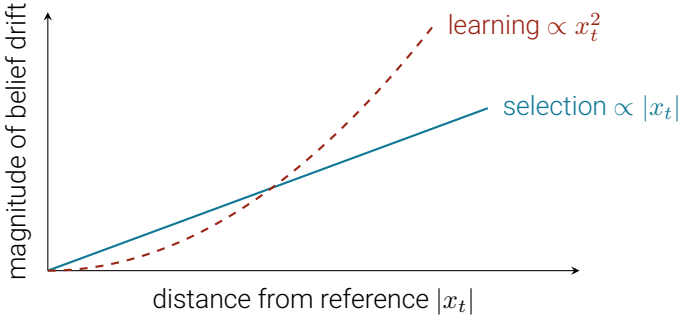
- Expected change in adult Romerian share:

$$\mathbb{E}_t[\phi_{t+1} - \phi_t] = \underbrace{\lambda \frac{\phi_t(1-\phi_t)(n_{R,t} - n_{M,t})}{\bar{n}_t}}_{\text{reproductive selection}} + \underbrace{(1-\lambda)(\pi_t - \phi_t)}_{\text{recruitment to credence}} + \underbrace{(1-\lambda)\mathbb{E}_t[\pi_{t+1} - \pi_t]}_{\text{new evidence}}$$

- Selection favors trajectory-optimists; learning favors the truth

Expected path	Malthusian world	Romerian world
Above reference	Selection: Romerians ×	Selection: Romerians ✓
	Learning: Malthusians ✓	Learning: Romerians ✓
Below reference	Selection: Malthusians ✓	Selection: Malthusians ×
	Learning: Malthusians ✓	Learning: Romerians ✓

Two Engines of Belief Change



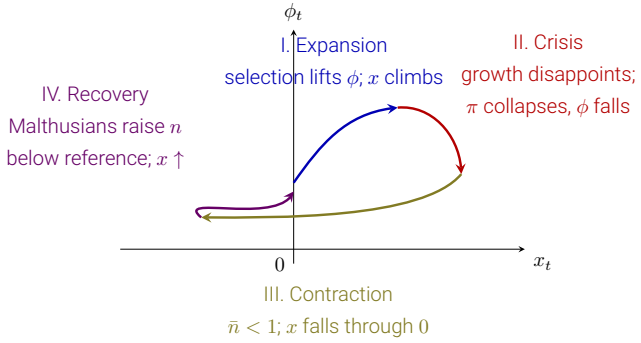
- Selection favors trajectory-optimists — vanishes linearly with $|x_t|$
- Learning favors the truth — vanishes quadratically
- Near reference: selection dominates; far away: learning dominates
- At $x = 0$: both exactly zero — disagreement is self-sustaining

Information Accumulation

- Cumulative signal information: $\mathcal{C}_T = \sum_{t=0}^{T-1} \mathcal{I}_t \propto \sum_{t=0}^{T-1} x_t^2$
- Theorem (Information and posterior convergence):
  - → On $\{\sum_t \mathcal{I}_t = \infty\}$: social credence $\pi_t \to$ truth a.s.
  - → On $\{\sum_t \mathcal{I}_t < \infty\}$: $\pi_t \to$ interior limit $\pi_\infty \in (0, 1)$
- Sustained fertility above or below replacement ⇒ gap stays away from zero ⇒ infinite info ⇒ truth identified
- Geometric return to reference ⇒ $\sum x_t^2 < \infty$ ⇒ finite info ⇒ persistent disagreement
- Whether learning resolves the debate depends on the population path

Malthusian World: Boom-Bust Cycles

cycle mechanics



- Selection and learning oppose above the reference $\Rightarrow$ tug-of-war
- Busts reinforced by selection; recoveries engineered by pessimists (they raise fertility when $x < 0$)
- Damped cycles: $\sum x_t^2 < \infty$ possible $\Rightarrow$ dormancy with accumulated belief

Romerian World: Dormancy and Escape

trap regimes

- Above reference: selection and learning aligned — both favor Romerians
  $\Rightarrow$ self-cementing escape and growth
- Below or near reference:
  - $\rightarrow$ Exact: $x_0 = 0 \Rightarrow x_t = 0, n_{R,t} = n_{M,t} = 1, \pi_t = \pi_0$ forever; wrong beliefs never corrected
  - $\rightarrow$ Near-dormant: for $|x_0|$ small, with high probability:

$$|x_t| \leq q^t |x_0|, \qquad \sum_t \mathcal{I}_t < \infty, \qquad \pi_t \to \pi_\infty \in (0,1)$$

Population converges to a finite level while leaving the scale-effect disagreement unresolved

Quantitative Findings

Application: Delayed Takeoff

- Setup: the dawn of the industrial revolution when history supports Malthusian beliefs (Clark 2007)
- Potential regime change (Hansen-Prescott 2002)
- Model: people might not be aware of it
  1. How long could the initial pessimism delay growth take-off?
  2. How large is the welfare consequences of imperfect information?
  3. How large is the role of luck or cultural entrepreneurship (Mokyr 2016)?

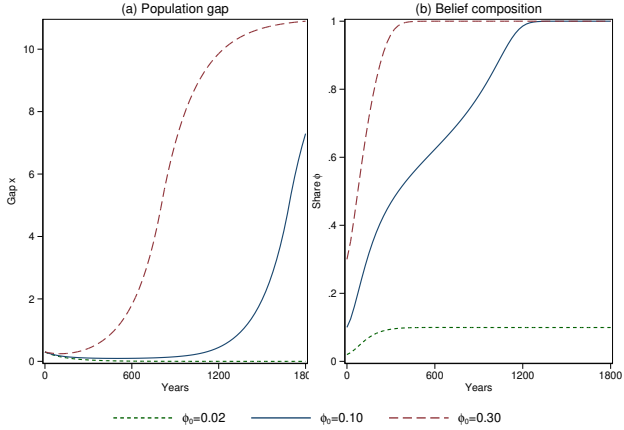
Calibration

computation details

Parameter	Value	Target / source
ρ, ε	0.50	CRRA & dynastic curvature (Jones–Schoonbroodt)
γ	0.312	15% child-rearing time at replacement
β	0.252	Replacement normalization $\Rightarrow \delta \simeq 0.253$
σ	0.20	≈4% annual s.d. of productivity
μ	0.10	Gap half-life ≈165 years (Jones 1995, 2022)
θ_R	0.50	Multiplier $\theta_R/\mu = 5$ (Kremer 1993); 2%/yr at entry
θ_M	-0.15	Perceived elasticity $-\theta_M/\mu = 1.5$ (Lee 1980, Clark 2007)
λ	0.60	Vertical transmission (Bisin–Verdier 2001, 2011)
x_0	0.30	Initial displacement

Threshold Dynamics

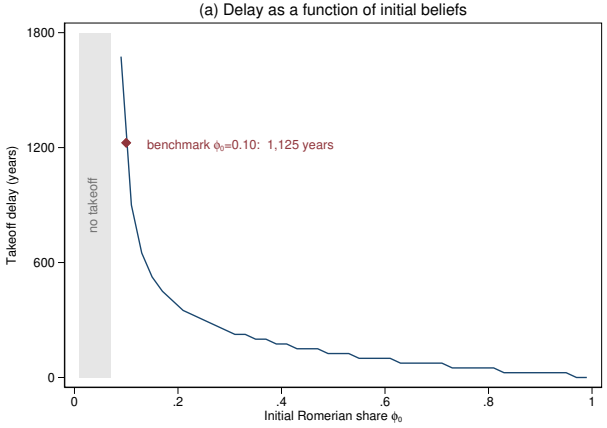
Monte Carlo | sensitivity



- $\phi_0 = 0.30$: fast takeoff (generation 20, ∼500 years)
- $\phi_0 = 0.10$: gap contracts to 0.09, composition crosses p^* at gen. 20; takeoff at gen. 55 (∼1,375 yrs)
- $\phi_0 = 0.02$: trapped — credence converges to < 1, gap returns to zero

Takeoff Delay and the Cost of Disagreement

welfare



- $\phi_0 \leq 0.07$: no takeoff within 2,000 years (numerical dormant trap)
- Delay falls steeply: 1,675 yrs at $\phi_0 = 0.09$; <300 yrs at $\phi_0 = 0.25$
- Welfare cost: $\omega \simeq 0.74\%$ of consumption per generation (20-gen window)

Conclusion

- Disagreement about the sign of population scale effects generates:
  - → state-dependent fertility rankings
  - → boom-bust cycles in Malthusian worlds
  - → information traps in in Romerian worlds
- Pessimism delays takeoff: inherited beliefs contract the gap and starve future generations of evidence

Ecological vs. Economic Malthusianism

back

- GSS question is in environment modules: captures ecological concern (resource depletion, sustainability)
- For the fertility margin, distinction is second-order: A_t is broad TFP, and both idioms point to $\theta < 0$, a poorer future for descendants
- Including a measure of general environmental concern leaves the Malthusian coefficient essentially unchanged
- For historical claims the distinction is more substantive; the survey only validates the belief-to-fertility arrow

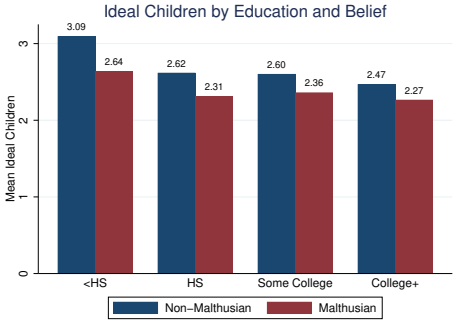
Regression Evidence: Continuous Scale and Counts

back

	Ideal Fertility	Actual Fertility
Malthusian (binary)	-0.268^{***}	-0.292^{***}
	(0.041)	(0.066)
Malthusian (1–5 scale)	-0.153^{***}	-0.146^{***}
	(0.021)	(0.033)
Poisson (binary)	-0.107^{***}	-0.179^{***}
	(0.016)	(0.037)
Excl. neutrals (binary)	-0.357^{***}	-0.342^{***}
	(0.054)	(0.087)
Controls + Year FE	Yes	Yes
Observations	1,822	1,822

Gradient Holds Within Education Groups

back



	Ideal		Actual	
	HS or Less	College+	HS or Less	College+
Malthusian	-0.318^{***}	-0.235^{***}	-0.209^{*}	-0.392^{***}
	(0.069)	(0.049)	(0.115)	(0.081)

Completed Fertility

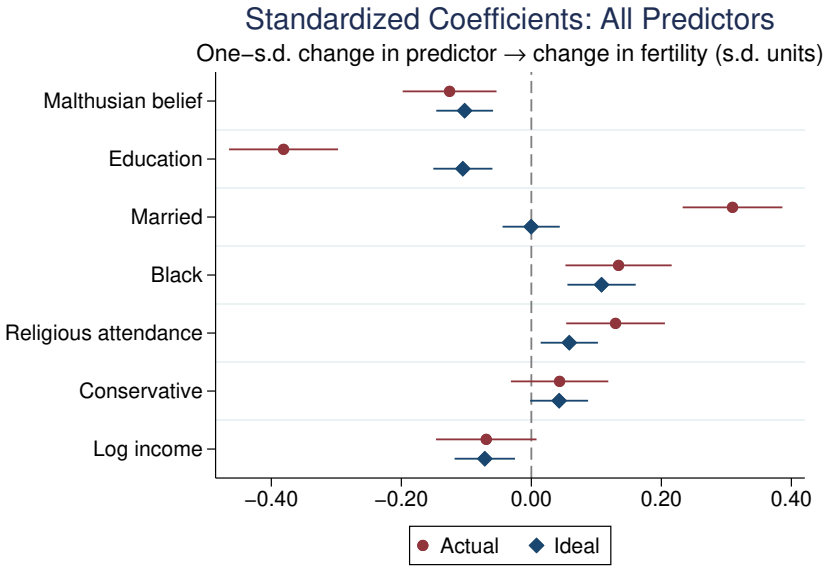
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	Full Sample	Completed	Not Yet Completed
Malthusian	-0.292^{***}	-0.257^{**}	-0.288^{***}
	(0.066)	(0.109)	(0.079)
Observations	1,822	848	974

- Completed = women 45+, men 50+; all controls included
- Estimate stable: not driven by incomplete childbearing among the young

Normalized Coefficients

back



Extensive vs. Intensive Margin

back

	Extensive: Any Children		Intensive: Children $\|> 0$	
	Baseline	Full	Baseline	Full
Logit (OR)	0.554^{***}	0.533^{***}		
	(0.067)	(0.080)		
AME	-0.092^{***}	-0.080^{***}		
	(0.019)	(0.019)		
OLS			-0.122^{*}	-0.085
			(0.072)	(0.080)
N	1,822	1,567	1,299	1,120

Association concentrated on the decision to become a parent.

Reverse Causality Checks

back

	Full Sample	Age 18–34	Age 35–49	Age 50+
Panel A: Ideal Fertility				
Malthusian	-0.268^{***}	-0.243^{***}	-0.322^{***}	-0.253^{***}
	(0.041)	(0.075)	(0.079)	(0.061)
Panel B: Actual Fertility				
Malthusian	-0.292^{***}	-0.052	-0.469^{***}	-0.263^{**}
	(0.066)	(0.094)	(0.124)	(0.114)
N	1,822	514	541	767

- Ideal gradient present even among young adults with little childbearing experience
- Actual: zero among 18–34, large among 35+ $\longrightarrow$ consistent with beliefs affecting fertility

Why Each Preference Ingredient Is Load-Bearing

back

- Time cost + CRRA: TFP level cancels; income-supports-children channel off by construction — beliefs enter only via growth forecasts
- $\rho < 1$: required for positive dynastic value; $\rho > 1$ (with $\varepsilon > 1$) flips gradient sign — rejected by data
- $\varepsilon \in (0, 1)$: total weight on children increases with n at diminishing rate; well-posed fertility problem
- Imperfect transmission: $\delta\lambda$ discounting in the fertility differential; association between parent's type and distant descendants decays
- Replacement normalization: $\beta^\dagger$ set so $n^0 = 1$; implied $\delta \simeq 0.253$

Microfoundation of the Population Gap

back

- Jones idea production: $\ln(A_{t+1}/A_t) = \nu + \lambda \ln N_t - \omega \ln A_t + \eta_{t+1}$; defining x_t appropriately yields $\mu = \omega\theta$
- Limit $\mu \to 0$ reproduces Kremer–Romer strong-scale-effects; trap reverts to slow learning
- Absorbing trap is a genuine semi-endogenous phenomenon
- θ encodes growth channel net of congestion; level channel neutralized by preferences
- The reference index H_t adjusts gradually: $\ln H_{t+1} - \ln H_t = \mu x_t$
- Convergence of the gap to zero can occur at a population level different from the initial value

Local Equilibrium Response: Details

back

- The local gap coefficient solves:

$$\big(a(p) - (1-\mu)\big)\big(1 - \delta a(p)\big) = K\bar{\theta}_p$$

- Explicit root: $a(p) = \frac{1+\delta(1-\mu)-\sqrt{[1-\delta(1-\mu)]^2-4\delta K\bar{\theta}_p}}{2\delta}$
- Individual policy slopes: $\left.\frac{\partial n_j}{\partial x}\right|_0 = b(p) + \frac{K(\theta_j - \bar{\theta}_p)}{1-\delta\lambda a(p)}$
- Amplification threshold: $a(p) = 1$ when $\bar{\theta}_p = \bar{\theta}^{dest} = \frac{\mu(1-\delta)}{K}$
- At baseline: $\bar{\theta}^{dest} \simeq 0.224$, requiring Romerian share $p^* \simeq 0.58$

Dormant Trap: Formal Statement

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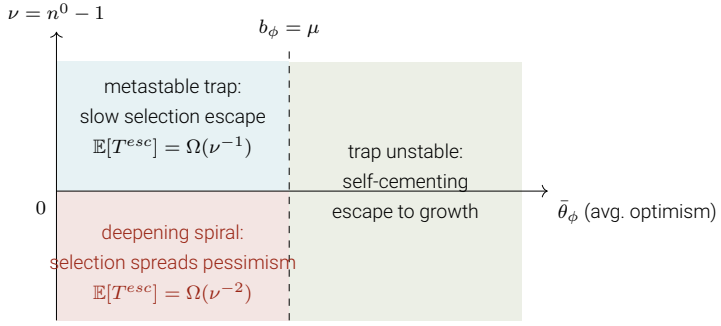
- Require uniform local return: $|F_x(0, \phi, \pi)| \leq q_0 < 1$ throughout a belief neighborhood
- On no-exit event E (probability $\geq 1 - C|x_0|/d$):

$$|x_t| \leq q^t |x_0|, \qquad \sum_{t=0}^{\infty} \mathcal{I}_t \leq \frac{\Delta\theta^2 x_0^2}{2\sigma^2(1-q^2)}$$

- Limiting credence is close to initial: $\mathbb{E}[\mathbb{1}_E |\pi_\infty - p|] \leq C|x_0|$
- With $\lambda < 1$: adult composition converges to π_∞
- Population converges to a positive finite level
- Long-run average log-productivity growth equals zero a.s. on E

Trap Regimes Diagram

back



- $\nu > 0$: selection slowly favors Romerians; eventual escape
- $\nu < 0$: selection favors Malthusians; escape time quadratic in $|\nu|$

Damped Cycles and Periodic Orbits

back

- Corollary (Damped demographic cycles): under uniformly negative $a(p) \in (-1, 0)$:

$$\operatorname{sgn}(x_{t+1}) = -\operatorname{sgn}(x_t), \qquad |x_t| \leq q^t |x_0|$$

  Population growth alternates; cumulative info finite; posterior interior

- Proposition (No periodic orbits under mean signals): replacing each signal by $s_t = \theta^* x_t$, no nonconstant periodic orbit with interior credence exists
  - → Every nonzero gap adds evidence for the truth
  - → Recurrent fixed-amplitude cycles require forgetting or outside shocks
- Damped-overshoot condition: $-\frac{(2-\mu)(1+\delta)}{K} < \bar{\theta}_p < -\frac{1-\mu}{K}$

Computation Details

back

- Fertility implements first-order log-form response:

$$\ln n_{j,t} = \nu + K\left[\bar{\theta}_p X_t^1 + (\theta_j - \bar{\theta}_p) X_t^\lambda\right]$$

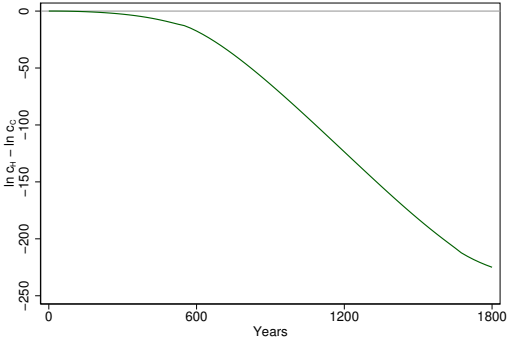
- X_t^λ, X_t^1 evaluated via constant-fertility forecast rule
- Aggregate fertility: within-period fixed point (local slope ≈ 0.07, tolerance 10^{-12})
- TFR range: $[0.6, 6]$; beliefs capped at $(10^{-9}, 1 - 10^{-9})$
- Modern θ_R remark: semi-endogenous estimates (Jones 2022, Bloom et al. 2020) imply $\theta \approx 0.03 \ll \bar{\theta}^{dest}$ — even correct beliefs would not amplify
- Historical calibration uses strong-scale regime of Kremer (1993)

Welfare Construction and Consumption Paths

back

Welfare measure:

- Dynastic consumption-equivalent over 20-generation window
- $\omega = (V_0^C / V_0^H)^{1/(1-\rho)} - 1$
- Benchmark: $\omega \simeq 0.74\%$ per generation
- Stable across $\lambda \times \sigma$ grid (0.71–0.75%)
- Conservative: $\delta \simeq 0.25$ per gen discounts distant future

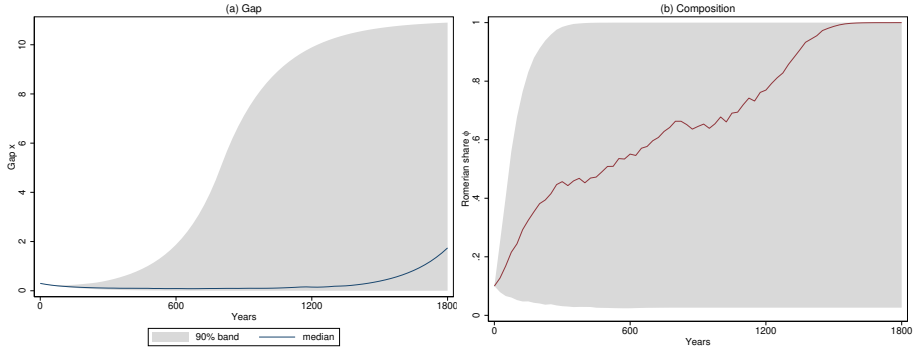


Levels beyond the takeoff of both economies reflect the fertility cap and are illustrative only.

Log-consumption difference $H - C$: positive early (fewer children $\Rightarrow$ more time), negative and widening thereafter.

Monte Carlo: The Role of Luck

back



- 500 replications, benchmark $\phi_0 = 0.10$; takeoff timing widely dispersed
- Fastest 5%: entry $<$500 years; bottom 5%: trapped forever
- The trap arises with positive probability even under the Romerian truth

Parameter Sensitivity

back

Parameter variation	Delay (years)	Mechanism
$\lambda = 0.4$	950	Faster recruitment to credence
$\lambda = 0.6$ (baseline)	1,125	—
$\lambda = 0.8$	>1,775	Family transmission insulates
$\sigma = 0.1$	175	4× more info per observation
$\sigma = 0.2$ (baseline)	1,125	—
$\sigma = 0.3$	>1,775	Too noisy to learn
$\theta_M = -0.075$	975	Weaker contraction but less info
$\theta_M = -0.15$ (baseline)	1,125	—
$\theta_M = -0.30$	1,225	Stronger contraction offsets more info

- Transmission and noise interact strongly: high λ + high $\sigma \Rightarrow$ censored
- θ_M moves delay modestly: contraction and info effects largely offset

Robustness: What Would Break the Results

back

- No common crossing: rival models always distinguishable $\Rightarrow$ learning always active; trap vanishes
- $\rho > 1$: all rankings reverse, gradient flips — rejected by data
- $\mu \to 0$: trap reverts to slow learning; absorbing character requires $\mu > 0$
- Vanishing noise: trap dissolves if $\sigma^2(x)$ shrinks as fast as the squared gap
- Outside evidence: regional comparisons or research can identify θ^* without domestic demographic departure
- Persistent demographic shocks: can repeatedly displace economy into informative states, restoring learning
- Type-specific preferences: would preserve fertility differences at the reference $\Rightarrow$ exact dormancy breaks